

Harmonic Balance Analysis of Nonlinear Dielectrics in Waveguide Devices

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Abstract—The computation of high order harmonic components of the time-harmonic field originated in waveguide structures containing nonlinear dielectrics is addressed exploiting Harmonic Balance and Finite Elements. The solution of the nonlinear system is handled either via Picard or relaxed iterations. Being the nonlinear loop particularly time demanding, a domain decomposition method is proposed to speed-up the system solution by restricting the iteration loop to a small sub-domain.

Index Terms—Harmonic balance (Harmonic analysis), finite element methods, domain decomposition method (si ritrova in diversi articoli), nonlinear media, waveguide filters

I. INTRODUCTION

TODAY's microwave and millimeter wave passive devices design requires, when high power densities are involved, the modeling of material nonlinear properties to accurately predict the steady state electromagnetic field.

Among the available techniques, the most immediate would be an explicit time-domain approach, which indeed often results to be computationally prohibitive [1]. An alternative is the time-harmonic approach, which can be based on the finite element (FE) solution of the frequency domain partial differential equation (PDE), obtaining the field quantities at a given frequency. Unfortunately, a single frequency is not sufficient to include all relevant phenomena, high order harmonics being generated from the interaction of the electromagnetic field with the nonlinear materials. Incorporating a larger set of harmonics in the FE formulation by using the Harmonic Balance Finite Element (HBFE) method [2], [3], also known as Multi-Harmonic Finite Element method [4]–[6], provides a fast and accurate solution to the problem as long as the harmonics required to effectively represent the field is not excessive.

The nonlinearity of the PDE is tackled by an iterative solution of a spatially and spectrally coupled large-scale linear system in which the nonlinear material properties are computed at each iteration on the basis of the previous iterations field values. Iteration stops when the computed solution converge, within a prescribed tolerance, to the sought solution. An algebraic Domain Decomposition technique as proposed in [7] improves computational timings, especially when the region occupied by nonlinear materials is rather small. But in [7] no Harmonic Balance was present and hence only the fundamental frequency was computed.

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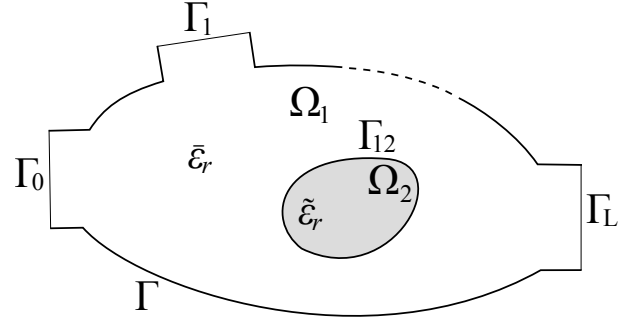


Fig. 1. Generic multiport device where the total domain Ω is subdivided into a large, linear subdomain Ω_1 and a smaller subdomain Ω_2 .

In this letter, we expand [7] by implementing a full harmonic balance domain decomposition finite element (HB-DDFE) method. To prove the capability and efficiency of the method the analysis of a waveguide filter containing dielectrics presenting a nonlinear, isotropic, permittivity is shown.

In following section II, the HBDDFE formulation is presented. In section III, simulation results for a broadband E-plane bandpass filter with a dielectric slab exhibiting a Kerr-type nonlinearity are shown, and the performances of the HBDDFE are discussed.

II. FORMULATION

The formulation given in this section holds for a generic multiport device as shown in Fig. 1. The domain Ω , with boundary Γ , can be either defined on $\mathbb{R}^2$ or $\mathbb{R}^3$. For the sake of simplicity, the examples presented in the next section are analyzed, without loss of generality, in $\mathbb{R}^2$. Ω is divided into a large part, Ω_1 , comprising only linear media, and a small part, Ω_2 , comprising all nonlinear media, with $\Omega = \Omega_1 \cup \Omega_2$ and $\Omega_1 \cap \Omega_2 = \emptyset$.

The permittivity in Ω can be expressed as

$$\epsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) = \begin{cases} \bar{\epsilon}_r(\mathbf{r}), & \mathbf{r} \in \Omega_1 \\ \tilde{\epsilon}_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) = \bar{\epsilon}_r(\mathbf{r}) + \mathcal{L}(\mathbf{E}(\mathbf{r})), & \mathbf{r} \in \Omega_2 \end{cases} \quad (1)$$

where $\mathbf{E}(\mathbf{r})$ is the electric field in the generic point $\mathbf{r}$ and $\mathcal{L}(\cdot)$ denotes a generic scalar operator describing the nonlinear behavior [7].

The electric field satisfies, within Ω , the vector Helmholtz equation

$$\nabla \times \left(\frac{1}{\mu_r} \nabla \times \mathbf{E}(\mathbf{r}) \right) - k_0^2 \epsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \mathbf{E}(\mathbf{r}) = 0 \quad (2)$$

with suitable boundary conditions on Γ , being k_0 the free space wavenumber. For the example shown in section III, Γ comprises a part Γ_0 which will be considered perfect electric conductor, and L parts Γ_l corresponding to the ports, where a modal expansion of the field is enforced [8].

In a standard FE Galerkin framework the domain Ω is discretized into a set of non overlapping finite elements and the electric field is approximated as a linear combination of vector basis functions α_n

$$\mathbf{E}(\mathbf{r}) = \sum_{n=1}^N E_n \alpha_n(\mathbf{r}) \quad (3)$$

Then by testing (2) over Ω with weighting functions α_m , chosen equal to the bases, the following weak form is obtained:

$$\begin{aligned} \int_{\Omega} \nabla \times \mathbf{W}_m(\mathbf{r}) \cdot \frac{1}{\mu_r} \nabla \times \mathbf{E}(\mathbf{r}) d\Omega \\ - k_0^2 \int_{\Omega} \mathbf{W}_m(\mathbf{r}) \cdot \varepsilon_r(\mathbf{r}, \mathbf{r}) \mathbf{E}(\mathbf{r}) d\Omega \\ + \oint_{\Gamma_0} \mathbf{W}_m(\mathbf{r}) \cdot \hat{n} \times \frac{1}{\mu_r} \nabla \times \mathbf{E}(\mathbf{r}) d\Gamma = 0 \end{aligned} \quad (4)$$

The multi-harmonic dependence in the HBFE method is introduced by approximating the electric field as [5]

$$\hat{\mathbf{E}}(\mathbf{r}, t) = \sum_{p=1}^P \mathbf{E}_p^{(s)}(\mathbf{r}) \sin(p\omega_0 t) + \mathbf{E}_p^{(c)}(\mathbf{r}) \cos(p\omega_0 t) \quad (5)$$

where ω_0 is the fundamental angular frequency and, here and in the following, the hat over a quantity denotes its approximated value given by the truncated Fourier series. The Fourier expansion is devoid of a dc-term as a static field does not represent a solution to (2) for $k_0 > 0$. $\mathbf{E}_p^{(s)}$ and $\mathbf{E}_p^{(c)}$ are the harmonic complex amplitudes that expand the field at $p\omega_0$ and hence they must individually satisfy (2) with $k_p = pk_0$. This is done in the FE framework by using the expansion (3) for each coefficient $\mathbf{E}_p^{(s)}$ and $\mathbf{E}_p^{(c)}$.

The scalar relative permittivity can also be approximated as a truncated Fourier series

$$\hat{\varepsilon}_r(\mathbf{E}(\mathbf{r}), \mathbf{r}, t) = \varepsilon_{r0} + \sum_{k=1}^K \varepsilon_{rk}^{(s)} \sin(k\omega t) + \varepsilon_{rk}^{(c)} \cos(k\omega t) \quad (6)$$

and the coefficients of the expansion are given by

$$\begin{aligned} \varepsilon_{r0} &= \frac{\omega}{2\pi} \int_0^{\frac{2\pi}{\omega}} \varepsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) dt \\ \varepsilon_{rk}^{(s)} &= \frac{\omega}{\pi} \int_0^{\frac{2\pi}{\omega}} \varepsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \sin(k\omega t) dt \\ \varepsilon_{rk}^{(c)} &= \frac{\omega}{\pi} \int_0^{\frac{2\pi}{\omega}} \varepsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \cos(k\omega t) dt \end{aligned} \quad (7)$$

The orders of the approximations P and K reflects on the accuracy of the solution and must be carefully chosen on the specific device analyzed and on the intensity of the incident field. These can be set automatically in the algorithm by selecting an appropriate convergence criterion on the power associated to the fundamental frequency as the number of harmonic coefficients is increased [9].

In a conventional, linear, FE approach discretization of (4) via (3) leads to a system:

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1N} \\ a_{21} & a_{22} & \dots & a_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ a_{N1} & a_{N2} & \dots & a_{NN} \end{bmatrix} \begin{bmatrix} E_1 \\ E_2 \\ \vdots \\ E_N \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_N \end{bmatrix} \quad (8)$$

where a_{ij} are built by assembling the integrals in (4) on an element-by-element basis [8]. In the present case, on the other hand, by substituting to $\mathbf{E}(\mathbf{r})$ and $\varepsilon_r(\mathbf{r})$ their respective harmonic approximations $\hat{\mathbf{E}}(\mathbf{r}, t)$ and $\hat{\varepsilon}_r(\mathbf{r}, t)$ in (4), and by performing a further testing by multiplying it by $\sin(p\omega t)$ or $\cos(p\omega t)$, $p = 1 \dots P$, and integrating over the time period of the fundamental $[0, \frac{2\pi}{\omega}]$ the following large linear system is obtained:

$$\begin{bmatrix} [A]_{11} & [A]_{12} & \dots & [A]_{1N} \\ [A]_{21} & [A]_{22} & \dots & [A]_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ [A]_{N1} & [A]_{N2} & \dots & [A]_{NN} \end{bmatrix} \begin{bmatrix} [E]_1 \\ [E]_2 \\ \vdots \\ [E]_N \end{bmatrix} = \begin{bmatrix} [b]_1 \\ [b]_2 \\ \vdots \\ [b]_N \end{bmatrix} \quad (9)$$

which has formally the same structure as (8) but in which the generic entries are themselves matrices in the form:

$$[A]_{mn} = \begin{bmatrix} a_{mn}^{(1s1s)} & a_{mn}^{(1s1c)} & \dots & a_{mn}^{(1sPc)} \\ a_{mn}^{(1c1s)} & a_{mn}^{(1c1c)} & \dots & a_{mn}^{(1cPc)} \\ \vdots & \vdots & \ddots & \vdots \\ a_{mn}^{(Pc1s)} & a_{mn}^{(Pc1c)} & \dots & a_{mn}^{(PcPc)} \end{bmatrix} \quad (10)$$

$$[E]_m = \begin{bmatrix} E_n^{(1s)} & E_n^{(1c)} & \dots & E_n^{(Pc)} \end{bmatrix}^T \quad (11)$$

$$[b]_n = \begin{bmatrix} b_n^{(1s)} & b_n^{(1c)} & \dots & b_n^{(Pc)} \end{bmatrix}^T \quad (12)$$

where superscript (ps) or (pc) indicates that the corresponding entry was computed by testing (4) with $\sin(p\omega t)$ or $\cos(p\omega t)$, respectively.

The product that occurs in the second integral of (4), between the approximated field and the permittivity, induces coupling between the harmonic components which reflects on non-diagonal matrices $[A]_{mn}$. However, as the PDE we seek to solve is free of first order time derivative, there is no coupling between $\mathbf{E}_n^{(ps)}$ and $\mathbf{E}_n^{(pc)}$. Thus, we may retain either the sine or the cosine harmonic function to excite the electromagnetic device and to express the multi-harmonic behavior of the field, halving in such a way the system dimensions and leading to an NP order system.

The nonlinearity is finally handled via a relaxed iteration. As a starting point a null electric field is assumed, $[E]^0$ and computing the pertinent system matrix $[A]^0 = [A_{([E]^0)}]$ and known term $[b]^0 = [b_{([E]^0)}]$ and the system $[A]^0 [E]^1 = [b]^0$ solved. The newly computed field $[E]^1$ can now be used to update system matrix and known term and carry on the following relaxed iteration:

$$[A]^{q-1} [E]^q = [b]^{q-1} \quad (13)$$

with

$$[A]^{q-1} = [A_{(\gamma[E]^{q-1} + (1-\gamma)[E]^{q-2})}] \quad (14)$$

$$[b]^{q-1} = [b_{(\gamma[E]^{q-1} + (1-\gamma)[E]^{q-2})}] \quad (15)$$

with $q \geq 2$ and $\gamma \in (0, 1]$. $\gamma = 1$ gives the standard Picard iteration [7]. Highly nonlinear materials and high intensity impressed fields may hamper the convergence of the nonlinear iteration, due to the oscillation of the solutions around the expected values. Lowering the value of γ has the effect to damp that oscillation, hence rendering the iteration more robust but inevitably reducing the convergence rate [10].

The process is repeated until the relative error between the updated solution and the previous one, in the sense of the Euclidean norm, is less than a prescribed value τ :

$$\frac{\|[E]^q - [E]^{q-1}\|}{\|[E]^q\|} \leq \tau \quad (16)$$

The order of the resulting HBFE system is NP . In practical analyses, this number can become considerable, and, since the solutions have to be refined iteratively, building at each step a new linearized system, computing time can become too high.

To enhance the computational efficiency, a substructuring domain decomposition approach is here proposed [7], [11]–[13]. The multi-harmonic unknowns vector $[E]$ defined by the HBFE method over Ω can be split into two vectors $[E^{(1)}]$ and $[E^{(2)}]$ containing the unknowns belonging to interior points in Ω_1 and Ω_2 , respectively. Unknowns belonging to the boundaries $\Gamma \cup \Gamma_{12}$ are placed in a third vector $[E^c]$. The HBFE system can then be recast in the following HBDDFE form:

$$\begin{bmatrix} [M^{(1)}] & 0 & [P^{(1)}] \\ 0 & [M^{(2)}]^{q-1} & [P^{(2)}]^{q-1} \\ [Q^{(1)}] & [Q^{(2)}]^{q-1} & [C]^{q-1} \end{bmatrix} \begin{bmatrix} [E^{(1)}]^q \\ [E^{(2)}]^q \\ [E^c]^q \end{bmatrix} = \begin{bmatrix} [b^{(1)}] \\ [b^{(2)}]^{q-1} \\ [b^c]^{q-1} \end{bmatrix} \quad (17)$$

where $[M^{(1)}]$ contains the HBFE coefficients of the linear system related to the unknowns $[E^{(1)}]$ fully internal to Ω_1 , hence with null harmonic coupling coefficients. $[M^{(2)}]$ contains coefficients for the nonlinear sub-domain Ω_2 . $[P^{(\cdot)}]$ and $[Q^{(\cdot)}]$ represent couplings between the interior unknowns and the ones on the boundaries which are collected in $[E^c]$.

By using the Schur complement concept, constructed with a Gauss elimination-type strategy, the boundary unknowns $[E^c]^q$ of (17), and hence the generalized scattering matrix of the device, can be retrieved [7], [13]

Hence, in order to solve the HBDDFE system and compute the steady state field, every single submatrix is assembled at first sight, then, within the iteration loop, only the submatrices related to Ω_2 and its boundary are to be updated. The efficiency of this DD technique relies on the fragmentation of the matrices and the subsequent computation of partial solutions. When the number of coefficients related to the nonlinear media is small, noticeable improvements can be achieved.

III. NUMERICAL RESULTS

In this section, the analysis of a millimeter wave broadband filter [14] is performed and the computational results are

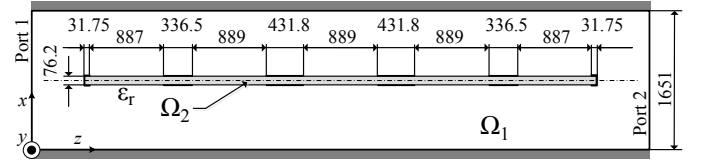


Fig. 2. Cross section of a broadband E-plane bandpass filter realized in an empty WR6 rectangular waveguide [14]. Measures are given in μm .

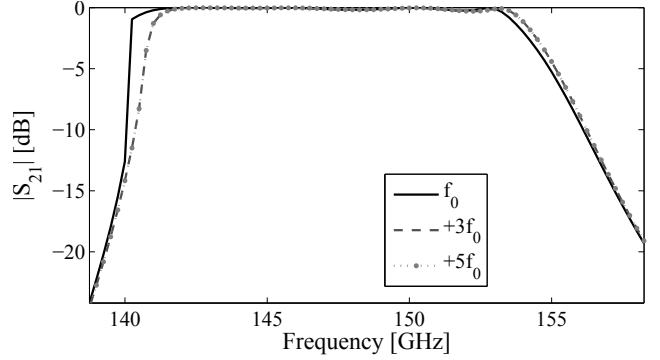


Fig. 3. Behavior of the TE_{10} spectral response as the number of harmonics retained for the HBFE increases ($E_i = 10\text{ kV/m}$).

shown. The filter is realized within a standard WR6 waveguide by properly designed metalizations on a $6040.5\text{ }\mu\text{m}$ long dielectric slab, mounted along the E-plane at the middle of the width range. A sketch of the H-plane cross section of the filter, plane of the performed 2D HBFE analysis, is shown in Fig. 2. First order nodal elements are used and the domain discretization is such that 2470 degrees of freedom are obtained, 42 belonging to the nonlinear subdomain Ω_2 , 456 to the boundaries and the remaining 1972 to the linear subdomain Ω_1 . The metalizations are considered perfectly conducting, thus imposing the appropriate Dirichlet conditions. The continuity of the field at the ports is imposed through a modal expansion which, for the geometry of the problem, consists of TE_{n0} modes, $n \in \mathbb{Z}$, where $n \leq 10$ have been chosen. The incident field at Port 1 is only of a TE_{10} mode and E_i is the associated electric field maximum amplitude.

For the example of section III, $\tau = 10^{-6}$ have been chosen.

The dielectric, enclosed in Ω_2 , presents a Kerr-type nonlinearity such that [15]

$$\tilde{\epsilon}_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) = \bar{\epsilon}_r(\mathbf{r}) + \alpha_2 |\mathbf{E}(\mathbf{r})|^2, \quad \mathbf{r} \in \Omega_2 \quad (18)$$

where $\bar{\epsilon}_r(\mathbf{r}) = 2.1$ and α_2 is chosen of $1.625 \cdot 10^{-10} \text{ m}^2/\text{V}^2$ according to [16]. The Kerr-like behavior of the permittivity induces the generation of odd order harmonics, thus the even ones have been neglected in the truncated Fourier expansion. Furthermore, as we choose a sinusoidal excitation, only $\sin(m\omega t)$ related coefficients of the electric field are retained.

We first analyze the effect of the number of harmonics retained on the filter's spectral response as shown in Fig. 3. For the nonlinearity degree represented by $E_i = 10 \text{ kV/m}$ and the chosen α_2 , the Picard iteration was sufficient to tackle the iteration loop. As we can see, the inclusion of the 3rd harmonic is crucial for proper bandwidth evaluation of the device. The field distribution within the device at 146 GHz, 3rd and 5th

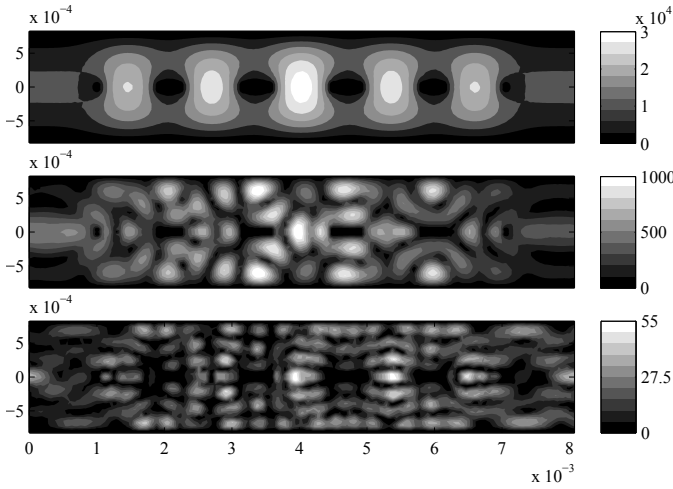


Fig. 4. $|E_y|$ distribution within the filter at fundamental ($f_0 = 146$ GHz), 3rd and 5th order harmonics ($E_i = 10$ kV/m).

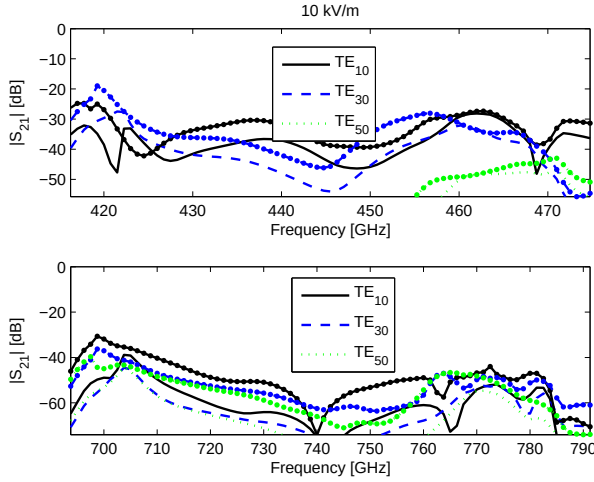


Fig. 5. GSM ... $E_i = 10$ kV/m.

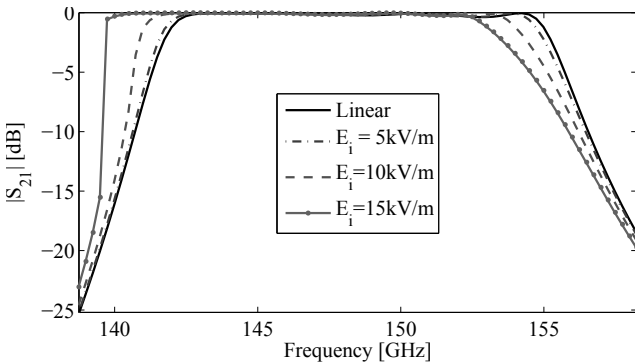


Fig. 6. S_{21} amplitude responses of the nonlinear filter given for several values of E_i compared to the linear case. The field is approximated up to the 5th harmonic.

harmonics are shown in Fig. 4.

To see whether the nonlinear dielectrics influences the spectral response of the device, the intensity of the field has been set to 5 kV/m, 10 kV/m and 15 kV/m corresponding to 18 mW, 72 mW and 162 mW of incident power, respectively,

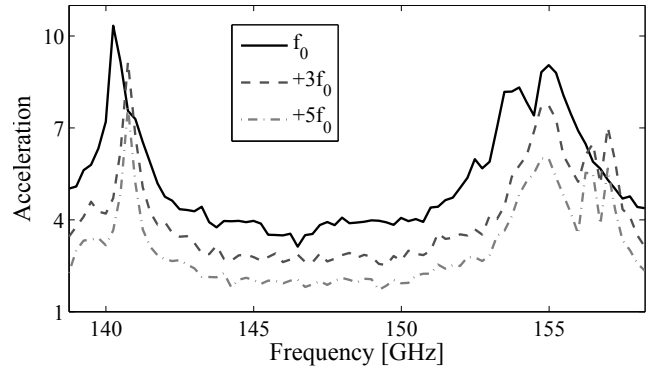


Fig. 7. Acceleration spectrum obtained for a HBFE systems of several harmonic orders. The computations are made for $E_i = 10$ kV/m.

which are common power values encountered in D-band applications [17]. The results, compared to those of a linear device ($\alpha_2 = 0$), are shown in Fig. 6. The field is approximated up to the 5th harmonic. There is an evident shift the bandpass range to lower frequencies as higher power densities are involved. $E_i = 15$ kV/m has required a relaxed iteration with $\gamma = 0.1$. In this case, the abrupt variation of $|S_{21}|$ around 139.5 GHz points out that the harmonic order used is not sufficient to accurately analyze the device. Furthermore, a finer mesh would be necessary to validate the inclusion of higher order harmonics.

The efficiency of the DD technique versus the full domain solution is analyzed introducing the *acceleration*

$$a \simeq \frac{I \cdot (T_\Omega)}{T_{\Omega_1} + I \cdot (T_{\Omega_2} + T_\Gamma)} \quad (19)$$

where I is the number of iterations required by the nonlinear loop and T_Ω , T_{Ω_1} , T_{Ω_2} and T_Γ are assembling and solving times. Fig. 7 depicts the spectral acceleration obtained increasing the number of harmonics for $E_i = 10$ kV/m. In this case, as stated before, at least a 3rd harmonic is required for accurate solution. In the average, higher order HBFE systems have lower acceleration, the matrices dimensions related to nonlinear media increasing. With a 3rd order system, the acceleration varies in the range [2.51, 9.15] such that the overall acceleration is of 4.18. In this case, the DD HBFE solution uses 24% of the time required by the full domain HBFE solution.

IV. CONCLUSION

An Harmonic Balance approach to the FE solution of nonlinear dielectrics filled waveguide devices has been presented and the accuracy of the multi-harmonic solution discussed. A domain decomposition technique has been used to speed-up the computations and the efficiency of the algorithm has been evaluated.

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